

Homework (Habanero)

- Q1.** What is the discriminant of the quadratic equation $x^2 + 5x + 6 = 0$?
(A) 1
(B) 5
(C) 13
(D) 17
(E) 25
- Q2.** The quadratic equation $x^2 - 7x + 12 = 0$ has how many solutions?
(A) 1
(B) 2
(C) 0
(D) 3
(E) 4
- Q3.** A patient is given a dosage of 0.5 mg of medicine per hour. The amount of medicine in the patient's system after t hours is given by the equation $A = 0.5t^2 + 2t + 1$. What is the discriminant of this quadratic?
(A) -3
(B) 1
(C) 4
(D) 7
(E) 12
- Q4.** A biologist is studying the growth of a cell culture. The population of the culture after x days is given by the equation $P = 2x^2 + 5x + 1$. How many solutions does this quadratic have?
(A) 1
(B) 2
(C) 0
(D) 3
- (E) 4
- Q5.** The quadratic equation $x^2 - 3x - 2 = 0$ has solutions of what type?
(A) Real and equal
(B) Real and distinct
(C) Complex and equal
(D) Complex and distinct
(E) Imaginary
- Q6.** What is the value of x in the quadratic equation $x^2 + 2x + 1 = 0$?
(A) -1
(B) 1
(C) -2
(D) 2
(E) 0
- Q7.** The quadratic equation $ax^2 + bx + c = 0$ has a discriminant of $b^2 - 4ac = 16$. What can be said about the solutions of this equation?
(A) They are real and equal
(B) They are real and distinct
(C) They are complex and equal
(D) They are complex and distinct
(E) They are imaginary
- Q8.** A quadratic equation has a discriminant of -7. What type of solutions does this equation have?
(A) Real and equal
(B) Real and distinct
(C) Complex and equal
(D) Complex and distinct
(E) Imaginary

Open Ended Questions (FRQ)

- Q9.** Solve the quadratic equation $x^2 + 4x + 3 = 0$ using the quadratic formula. Show all work and explain your answer.
- Q10.** A medical researcher is studying the effects of a new medicine on a patient. The amount of medicine in the patient's system after t hours is given by the equation $A = t^2 - 2t + 1$. Find the time at which the amount of medicine in the patient's system is a maximum. Show all work and explain your answer.

Chef's Tips

- Emphasize the importance of checking the discriminant before solving a quadratic equation
- Make sure students understand that the quadratic formula can be used to solve any quadratic equation
- Remind students to show all work and explain their answers when solving free-response questions